Matrix product formulas

Let $A = [a_{i\ell}] \in \mathbb{R}^{m\times n}$ and $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_k] \in \mathbb{R}^{n\times k}$.

Dimensions: the number of columns of A must equal the number of rows of B. The product has size $m \times k$:

$$(m \times n)(n \times k) \longrightarrow m \times k.$$

Column definition: apply A to each column of B:

$$AB = [A\mathbf{b}_1 \ A\mathbf{b}_2 \ \cdots \ A\mathbf{b}_k].$$

Entry theorem: row i of A dotted with column j of B, for $1 \le i \le m$, $1 \le j \le k$:

$$(AB)_{ij} = \sum_{\ell=1}^{n} a_{i\ell}b_{\ell j} = (a_{i1}, \ldots, a_{in}) \cdot (b_{1j}, \ldots, b_{nj}).$$

Matrix-vector special case: view $\mathbf{x} \in \mathbb{R}^n$ as an $n \times 1$ matrix. Then $A\mathbf{x}$ has size $m \times 1$ and represents a vector in $\mathbb{R}^m$.

Properties of multiplication

For matrices of compatible sizes in each identity and $r \in \mathbb{R}$:

Associativity: $A(BC) = (AB)C$.

Distributivity: $(A + B)C = AC + BC$ and $C(A + B) = CA + CB$.

Scalar multiplication: $A(rB) = r(AB) = (rA)B$.

Transpose of a product: $(AB)^T = B^T A^T$; the order reverses.

Order and commutativity

Order matters: AB existing does not imply that BA exists. Even when both exist, their sizes can differ, and equality is not guaranteed.

Commute: square matrices A, B of the same size commute when $AB = BA$. Matrix multiplication is not commutative in general.

Commuting cases: two diagonal matrices of the same size commute; every square matrix commutes with itself.

Composition of matrix maps

For $B \in \mathbb{R}^{n\times k}$ and $A \in \mathbb{R}^{m\times n}$,

$$T_B : \mathbb{R}^k \to \mathbb{R}^n, \qquad T_A : \mathbb{R}^n \to \mathbb{R}^m.$$

Composition rule: apply B first, then A. The composite map is $T_A \circ T_B : \mathbb{R}^k \to \mathbb{R}^m$, with matrix AB:

$$(T_A \circ T_B)(\mathbf{x}) = T_A(T_B(\mathbf{x})) = A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Reading the product: in $(AB)\mathbf{x}$, the rightmost matrix acts first.